

MCA-2006 - EXAMINATION PAPER

Directions for Q. (1-2) :

Each question below has one or two blanks indicating that something has to be filled in. Choose the appropriate pair of words from the given choices to fill up the blanks for the following questions :

1. Non-violent demonstrations often create such tensions that a community that has constantly refused to its injustices is forced to correct them: the injustices can no longer be
 1) acknowledge, ignored
 2) decrease, verified
 3) tolerate, accepted
 4) address, eliminated
2. Winsor McCay, the cartoonist, could draw with incredible : his comic strip about Little Nemo was characterised by marvelous draftsmanship and sequencing
 1) Sincerity 2) Efficiency
 3) Virtuosity 4) Rapidity

Directions for Q. (3-5) :

Each of the questions below consists of a word in capital letters followed by four words. Choose the one that is most nearly similar in meaning to the word in capital letters :

3. **NYMPH**
 1) Clinch 2) Fairy
 3) Flinch 4) Airy
4. **EMANATE**
 1) Originate 2) Terminate
 3) Produce 4) Identify
5. **INTERIM**
 1) Interval 2) Corner
 3) Internal 4) Temporary

Directions of Q. (6-8) :

Each question below consists of a word printed in capital letters, followed by four words. Choose the word that is most

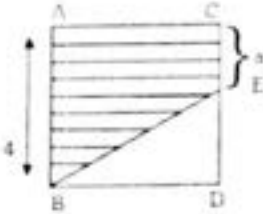
nearly opposite in meaning to the word in capital letters.

6. **BOYCOTT**
 1) Underwrite 2) Underbid
 3) Stipulate 4) Patronize
7. **ADULTERATION**
 1) Consternation 2) Purification
 3) Normalization 4) Approximation
8. **INSIPID**
 1) Tasty 2) Silly
 3) Angry 4) Scrap

Directions for Q. (9-11)

Each of the questions below consists of a pair of words in capital letters followed by four pair of words. Choose the pair of words which express the relationship similar to that expressed in the words in capital letters.

9. **CRIME : ACCOMPLICE**
 1) Legislation : Parliament
 2) Repair : Mechanic
 3) Journey : Train
 4) Text book : Co-author
10. **WEALTH : SQUANDER**
 1) Energy : Dissipate
 2) Power : Conserve
 3) Money : Make
 4) Gas : Produce
11. **CARICATURE : PORTRAIT**
 1) Cartoon : Drawing
 2) Mischief : Normal
 3) Comics : Happy
 4) Critical : Memorable
12. **Earthquake waves travel fastest in**
 1) Soil 2) Molten rock
 3) Water 4) Flexible rock

13. Which antibiotic can cure Anthrax, if detected at an early stage?
 1) Amoxycillin 2) Erythromycin
 3) Cephalosporin 4) Ciprofloxacin
14. The instrument 'Seismograph' is used to record
 1) Temperature of the sun
 2) Earthquake shock
 3) Pressure of air
 4) Accurate time
15. Which is the first letter to the right of the letter, which is immediately before the ninth letter from the left in the following alphabet?
 ABCDEFGHIJKLMNOPQRSTUVWXYZ
 1) R 2) T
 3) S 4) I
16. If $\frac{a}{b} = \frac{4}{3}$, then $\frac{3a+2b}{3a-2b} = ?$
 1) -1 2) 3
 3) 5 4) 6
17. Let n be any positive integer. Then $n^3 - n$ is divisible
 1) always by 12 2) never by 12
 3) always by 6 4) never by 6
18. Ten 50 paise coins are placed on a table. Six of these show tails while four show heads. A coin is chosen at random and flipped over (not tossed). This operation is performed seven times. One of the coins is then covered. Of the remaining nine coins five show tails and four show heads. The covered coin shows
 1) a head
 2) a tail
 3) more likely a head
 4) more likely a tail
19. The number of positive integers not greater than 100, which are not divisible by 2, 3 or 5 is
 1) 24 2) 26
 3) 29 4) None of these
20. Raman's salary was decreased by 50% and subsequently increased by 50% he has a loss of
 1) 0% 2) 25%
 3) 0.25% 4) 2.5%
21. The length of the ladder equals exactly the height of the wall. If the ladder is placed on a 2 feet tall stool placed 10 feet away from the wall then its tip can just touch the top of the wall. The height of the wall, in feet, is
 1) 15 2) 26
 3) 28 4) 32
22. Gopal went to a fruit market with certain amount of money. With this money he can buy either 50 oranges or 40 mangoes. He retains 10% of the money for taxi fare. If he buys 20 mangoes, then the number of oranges he can buy is
 1) 9 2) 18
 3) 20 4) 5
23. $A = \{0\}$, then the elements of $[A \times A]$ is
 1) ϕ 2) α
 3) β 4) $\{0,0\}$
24. Two numbers are respectively 30% and 40% more than the third number. What % is the first number of the second?
 1) 926 2) $92\frac{6}{7}$
 3) 70 4) $40\frac{5}{4}$
25. In the square below, whose side is 4 units, the ratio of the area of shaded portion to the area of unshaded region is

 1) $(4+a)/(4-a)$ 2) $(2+a)/(2-a)$
 3) $4a-3$ 4) $8a-4$
26. A spent Rs.1,000 to improve the fuel efficiency of his car from 12 km/litre of fuel to 13.5 km/litre. If the fuel cost is Rs.18 per litre, how many km will the car has to run in order for A to get back the amount spent in the form of fuel savings?
 1) 11666 kms 2) 7833 kms
 3) 6000 kms 4) 14000 kms

27. X travels from Koval to Tirupur at a speed of 80 kmph. He travelled back at a speed of 70 kmph. His average speed for the entire journey was
 1) 72.33 kmph 2) 75.5 kmph
 3) 75 kmph 4) 74.66 kmph
28. The sum of three consecutive odd number is 57. The middle one is
 1) 17 2) 19
 3) 21 4) 23
29. Of the two numbers 10^{11} and 11^{10} which is greater
 1) 10^{11}
 2) 11^{10}
 3) both are equal
 4) cannot be determined
30. Four small discs, each of radius 5 cms, are cut from a circular disc of radius 20 cms. The ratio of the left out area to the cut out area is
 1) 2 : 1 2) 3 : 1
 3) 4 : 1 4) 1 : 1
31. When the price of the Maruti cars increased by 30%, the number of Maruti cars sold decreased by 20%. The total sales revenue
 1) decreased by 20%
 2) increased by 10%
 3) increased by 4%
 4) remains same
32. The capacity of a tank of dimensions (8m $\times$ 6m $\times$ 2.5 m) is
 1) 120 litres 2) 1200 litres
 3) 12000 litres 4) 120000 litres
33. If x and y are different integers, both divisible by 5, then which of the following is not necessarily true?
 1) (x-y) is divisible by 5
 2) (x+y) is divisible by 10
 3) (x-y) is divisible by 10
 4) (x+y) is divisible by 5
34. Santhosh enters a positive integer y in the electronic calculator and then goes on pressing square root ($\sqrt{\quad}$) key repeatedly. Then
 1) the display shows error message
 2) the display becomes closer to 0
 3) the display becomes closer to 1
 4) none of the above is necessarily true and the answer depends on the choice of y
35. Let $x < 0$, $0 < y < 1$, $z > 1$. Which of the following is false?
 1) $(x^2 - z^2)$ has to be positive
 2) yz can be less than one
 3) xy can never be 0
 4) $(y^2 - z^2)$ is always negative
36. A report consists of 20 sheets each of 55 lines and each such line consists of 65 characters. This report is retyped on to sheets each of 65 lines such that each line consists of 70 characters. The percentage reduction in number of sheets is closest to
 1) 15 2) 20
 3) 35 4) 30
37. In a road network covering 10 cities, city C can be reached only from city A or city B. The distance from A to C is 50 kms and from B to C is 25 kms. The shortest distance from A to B is 40 kms, from city P to A is 200 kms, and from city P to B is 230 kms. Then the shortest distance from city P to city C is
 1) 225 kms 2) 265 kms
 3) 255 kms 4) 250 kms
38. The average weight of three men A, B and C is 84 kg. Another man D joins the group and the average now becomes 80 kg. If another man E, whose weight is 3 kg more than that of D, replaces A, then the average weight of B, C, D and E becomes 79 kg. The weight of A is
 1) 70 kg 2) 72 kg
 3) 75 kg 4) 80 kg
39. How many times do the hands of a clock point the same direction in a day?
 1) 12 2) 20
 3) 22 4) 24
40. The cost of making an article is divided between materials, labour and overheads in the ratio of 3 : 4 : 1. If the material cost is Rs.67.50, the cost of article in Rs. is
 1) 180 2) 122.50
 3) 380 4) 540
41. Which of the following comes next in the series below:
 nsi . org pqe qpc
 1) rqd 2) pqa
 3) aor 4) roa

42. Which of the following, given in the answers will logically fit into the next figure in the following sequence?

V 5 C ?

- 1) 50 2) 15
3) 100 4) 1000

Directions for Q. (43-44) :

Coach Balkisher is trying to put together a team of four players for a tennis tournament. He has seven players available : Males A, B and C and the females M, N, O and P. All players selected should have equal ability and they must be able to play with each other player. Player B cannot play with player M. Player C cannot play with player P. Player M cannot play with player O.

43. If player O is selected and player B is rejected the team will consist of

- 1) A, C, M, O 2) A, C, N, O
3) A, C, P, O 4) A, N, P, O

44. If player, M is in the team, which other players must be in the team as well

- 1) A, B, N 2) A, C, N
3) A, C, O 4) A, C, P

45. The next number in the series 64, 125, 216 is

- 1) 315 2) 343
3) 361 4) 400

46. What is the next number in the series? 3, 3, 3, 9, 15, 27, 51, 93,

- 1) 171 2) 147
3) 285 4) 483

47. Buffer is a

- 1) Permanent memory
2) Temporary memory
3) Screen memory
4) Parameter

48. OCR stands for

- 1) Outsized Character Reader
2) Optical Character Recognition
3) Operational Character Reader
4) Only Character Recognition

49. Data transfer rate in modems is measured in

- 1) bits / minute 2) bits / second
3) bytes / second 4) none of the above

50. The transformation of a message into an encoded form is known as

- 1) Mnemonics 2) Encoding
3) Encryption 4) Decoding

51. A way of condensing programs to pack most instructions into the smallest space is known

- 1) Crunching 2) Control panel
3) Concatenation 4) Cybernation

52. Time during which a job is processed by a computer is

- 1) Delay time 2) Execution time
3) Real time 4) Down time

53. What value does the term 4.5E4 represent in BASIC?

- 1) 450000.0 2) 45000.0
3) 4.500 4) 45.0000

54. What is the output of the following program?

```
#include "stdio.h"
main ()
```

```
{
    putchar('A');
    putchar('B');
    putchar('C');
}
    putchar(char ch)
{
    printf("%c", ch);
}
```

- 1) null 2) C
3) A 4) ABC

55. Convert Hex OXFEDB to Octal

- 1) 177333 2) 177332
3) 188330 4) None of the above

56. LAN stands for

- 1) Land Area Network
2) Local Area Network
3) Large Area Network
4) None of the above

57. Which of the following is not an input device?

- 1) Keyboard 2) Printer
3) Mouse 4) Light Pen

58. Stack is

- 1) a library function 2) a compiler
3) first in first out 4) last in first out

59. **MODEM is**

- 1) Modern mathematics
- 2) Modulator and demodulator
- 3) Modulating mathematics
- 4) None of the above

60. Which of the following is a solution to $x+x^2=1$?

- 1) -1
- 2) 1/2
- 3) 1
- 4) None of the above

61. If the average of 5 consecutive integers is 12 what is the sum of the least and greatest of the 5 integers?

- 1) 24
- 2) 14
- 3) 12
- 4) 11

62. From the following figure the value of $(x-y)/(x+y)$ is



- 1) 5/24
- 2) 1/4
- 3) 7/16
- 4) 11/24

63. If $xy \neq 0$, $(x-1)/xy = ?$

- 1) $1/x - 1/xy$
- 2) $x/y - 1/xy$
- 3) $1/y - x$
- 4) $1/y - 1/xy$

64. The number 0.01 is how many times as great as the number $(0.0001)^{2/3}$?

- 1) 10^2
- 2) 10^4
- 3) 10^6
- 4) 10^{10}

65. A certain cake recipe states that the cake should be baked in a pan 20 cm in diameter. If Jules wants to use the recipe to make a cake of the same depth but 30 cm in diameter by what factor should he multiply the recipe ingredients?

- 1) $2\frac{1}{2}$
- 2) $2\frac{1}{4}$
- 3) $1\frac{1}{2}$
- 4) $1\frac{4}{9}$

66. If $x > 0$ and $y > 0$, which of the following is equivalent to $(x/y) \sqrt{(y/x^2)}$?

- 1) $\sqrt{(x/y)}$
- 2) $\sqrt{(x)}$
- 3) $\frac{1}{\sqrt{x}}$
- 4) $\frac{1}{\sqrt{y}}$

67. If $F(x) = \int_1^x \log t \, dt$ for all positive x , then $F'(x) =$

- 1) x
- 2) $1/x$
- 3) $\log x$
- 4) $x \log x$

68. Consider the following sequence of instructions :

- I. Set $k=999$, $i=1$, and $p=0$
- II. If $k > i$ then go to step III, otherwise go to step V
- III. Replace i with $2i$ and replace p with $p+1$
- IV. Go to step II
- V. Print P

If these instructions are followed what number will be printed at step V?

- 1) 1
- 2) 2
- 3) 10
- 4) 512

69. If S is a nonempty finite set with K elements, then the number of one-to-one functions from S onto S is

- 1) $K!$
- 2) K^2
- 3) K^K
- 4) 2^K

70. A drawer contains 2 blue, 4 red and 2 yellow socks. If 2 socks are to be randomly selected from the drawer what is the probability that they will be of the same color?

- 1) $2/7$
- 2) $2/5$
- 3) $3/7$
- 4) $1/2$

71. When 20 children in a classroom line up for lunch, Pat insists on being somewhere ahead of Lynn. If Pat's demand is to be satisfied, in how many ways can the children line up?

- 1) $20!$
- 2) $19!$
- 3) $18!$
- 4) $20!/2$

72. How many integers from 1 to 1000 are divisible by 30 but not by 16?

- 1) 29
- 2) 31
- 3) 32
- 4) 33

73. Which of the following integrals on the interval $[0, \pi/4]$ has the greatest value?

- 1) $\int_0^{\pi/4} \sin t \, dt$
- 2) $\int_0^{\pi/4} \cos t \, dt$
- 3) $\int_0^{\pi/4} \cos^2 t \, dt$
- 4) $\int_0^{\pi/4} \cos 2t \, dt$

74. A fair coin is tossed 8 times. What is the probability that more of the tosses will result in heads than will result in tails?
- 1.3
 - 87.256
 - 23.64
 - 93.256
75. The inside of a certain water tank is a cube measuring 10 metre of each edge and having vertical sides and no top. Let $h(t)$ denotes the water level, in feet, above the floor of the tank at time t seconds. Starting at time $t=0$, water pours into the tank at a constant rate of 1 cubic metre per second and simultaneously water is removed from the tank at a rate of $0.25 h(t)$ cubic metre per second. As $t \rightarrow \infty$, what is the limit of the volume of the water in the tank?
- 400 cubic metre
 - 600 cubic metre
 - 1,000 cubic metre
 - The limit does not exist
76. What is the greatest integer that divides P^4-1 for every prime number p greater than 5?
- 30
 - 48
 - 120
 - 240
77. At how many points in the xy -plane do the graphs of $y=x^{12}$ and $y=2^x$ intersect?
- None
 - One
 - Two
 - Three
78. In a certain year, Salem produced 2.3 and Bhilal produced 1.6 of all the iron ore in India. If all the other states combined produced 18 million tons that year, how many million tons did Salem produce that year?
- 27
 - 36
 - 54
 - 72
79. The operation denoted by the symbol Δ is defined for all real number p and r as follows:
 $P \Delta r = pr - p + r$
 What is the value of $(-4) \Delta 5$?
- 9
 - 11
 - 19
 - 19
80. A certain software product has a mean time between failures of 10,000 hours and has a mean time to repair of 20 hours. If the product is used by 100 customers, what is its availability?
- 80%
 - 90%
 - 98%
 - 99.8%
81. Which of the following algorithms has running time $\theta(n^2)$ in the worst case but $\theta(n \log n)$ on average?
- Bubble sort
 - Merge sort
 - Heap sort
 - Quick sort
82. Which of the following is the name of the data structure in a compiler that is responsible for managing information about variables and their attributes?
- Abstract Syntax Tree
 - Attribute Grammar
 - Symbol Table
 - Semantic Stack
83. Consider the following pseudocode
- ```

x := 1; i := 1;
while (x ≤ 1000)
begin
 x := 2x;
 i := i + 1;
end ;

```
- What is the value of  $i$  at the end of the pseudocode?
- 4
  - 5
  - 6
  - 7
84. Consider the following pseudo code, where  $n$  is a non-negative integer
- ```

x := 0;
i := 0;
while i < n do
    x := x + 2i;
    i := i + 1;
end
  
```
- Which of the following is a loop invariant for the while statement?
- $x = 2^{i-1}$ and $0 \leq i < n$
 - $x = 2^{i+1} - 1$ and $0 \leq i < n$
 - $x = 2^i - 1$ and $0 \leq i \leq n$
 - $x = 2^{i+1} - 1$ and $0 \leq i \leq n$

85. Which of the following statements about a remote procedure call is true?

- 1) It is used to call procedures with addresses that are farther than 64 K bytes away
- 2) It cannot return a value
- 3) It cannot pass parameters by reference
- 4) It cannot call procedures implemented in a different language

86. Company X shipped 5 computer chips, 1 of which was defective and Company Y shipped 4 computer chips, 2 of which were defective. One computer chip is to be chosen uniformly at random from the 9 chips shipped by the companies. If the chosen chip is found to be defective, what is the probability that the chip came from Company Y?

- 1) $\frac{2}{9}$
- 2) $\frac{4}{9}$
- 3) $\frac{1}{2}$
- 4) $\frac{2}{3}$

87. The object-oriented paradigm includes which of the following properties

- I. Encapsulation
- II. Inheritance
- III. Recursion

- 1) I only
- 2) II only
- 3) I and II only
- 4) II and III only

88. If $A = (-6) 4$ and $B = (-6) 5$ then

- 1) $A > B$
- 2) $B > A$
- 3) $A = B$
- 4) Cannot be determined from the information given

89. If $A = \frac{4}{100}$ and $B = \frac{0.012}{3}$ then

- 1) $B > A$
- 2) $A = B$
- 3) $A > B$
- 4) Relationship indeterminate

90. If $x + 2y > 8$, $A = 2x + 4y$ and $B = 20$

- 1) Quantity A is greater
- 2) Quantity B is greater
- 3) Quantity A equals Quantity B
- 4) The relationship cannot be determined from the information given

Directions for Q. (91-93) :

Answer the questions by reading the passage below:

Three men (Tom, Peter and Jack) and three women (Eliza, Anne and Karen) are spending a few months at a hillside. They are to stay in a row of nine cottages, each one living in his or her own cottage. There are no others staying in the same row of cottages.

I. Anne, Tom and Jack do not want to stay in any cottage, which is at the end of the row.

II. Eliza and Anne are unwilling to stay besides any occupied cottage.

III. Karen is next Peter and Jack.

IV. Between Anne and Jack's cottage there is just one vacant cottage.

V. None of the girls occupy adjacent cottages.

VI. The cottage occupied by Tom is next to an end cottage.

91. Which of the above statements can be said to have been derived from two other statements?

- 1) Statement I
- 2) Statement II
- 3) Statement III
- 4) Statement V

92. How many of them occupy cottages next to a vacant cottage?

- 1) 2
- 2) 3
- 3) 4
- 4) 5

93. Which among these statement(s) are true?

(i) Anne is between Eliza and Jack

(ii) At the most four persons could have occupied cottages on either side of them

(iii) Tom stays besides Peter

- 1) (i) only
- 2) (ii) only
- 3) (i) and (iii) only
- 4) (ii) and (iii) only

Directions for Q. (94-97) :

Answer questions using the paragraph below:

Six scientists A, B, C, D, E and F are to present a paper each at a one-day conference. Three of them will present their papers in the morning session before the lunch break whereas the other three will be presenting in the afternoon session. The lectures have to be scheduled in such a way that they comply with the following restrictions:

- B should present his paper immediately before C's presentation, their presentations cannot be separated by the lunch break.
- D must be either the first or the last scientist to present his paper.

94. In case C is to be the fifth scientist to present his paper, then B must be

- 1) First
- 2) Second
- 3) Third
- 4) Fourth

95. B could be placed in any of the following places in the order of presenters EXCEPT

- 1) First
- 2) Second
- 3) Third
- 4) Fourth

96. In case F is to present his paper immediately after D presents his paper, C could be scheduled for which of the following places in the order of presenters?

- 1) Second
- 3) Fourth

- 2) Third
- 4) Fifth

97. In case F and E are the fifth and sixth presenters respectively then which of the following must be true?

- 1) A is first in the order of presenters
- 2) A is third in the order of presenters
- 3) A is fourth in the order of presenters
- 4) B is first in the order of presenters

98. A rectangle is 14 cm long and 10 cm wide. If the length is reduced by x cms and its width is increased also by x cms so as to make it a square then its area changes by

- 1) 4
- 2) 144
- 3) 12
- 4) 2

99. If 1 cm on a map corresponds to an actual distance of 40 kms and the distance on the map between Bombay and Calcutta is 37.5 cms, the actual distance between them is

- 1) 375 kms
- 2) 3750 kms
- 3) 1500 kms
- 4) 1375 kms

100. If $3/p=6$ and $3/q=15$ then $p-q=?$

- 1) $1/3$
- 2) $2/5$
- 3) $3/10$
- 4) $5/6$

ANSWERS

1.....3	2.....3	3.....2	4.....1	5.....4	6.....4	7.....2	8.....1	9.....4	10.....1
11.....2	12.....3	13.....3	14.....2	15.....4	16.....2	17.....3	18.....2	19.....2	20.....2
21.....2	22.....3	23.....4	24.....2	25.....1	26.....3	27.....4	28.....2	29.....1	30.....1
31.....3	32.....4	33.....2	34.....3	35.....1	36.....2	37.....4	38.....3	39.....3	40.....1
41.....4	42.....3	43.....2	44.....2	45.....2	46.....1	47.....2	48.....2	49.....4	50.....1
51.....1	52.....2	53.....2	54.....1	55.....1	56.....2	57.....2	58.....3	59.....2	60.....4
61.....1	62.....2	63.....4	64.....3	65.....2	66.....4	67.....3	68.....4	69.....1	70.....1
71.....3	72.....1	73.....2	74.....4	75.....2	76.....4	77.....3	78.....4	79.....2	80.....4
81.....2	82.....4	83.....4	84.....2	85.....1	86.....1	87.....4	88.....1	89.....3	90.....2
91.....4	92.....2	93.....3	94.....4	95.....3	96.....4	97.....3	98.....1	99.....3	100.....3